

MAC sublayer (Medium Access Control)

Multiple access protocols

Their job is to control how multiple devices share the same communication medium without colliding.

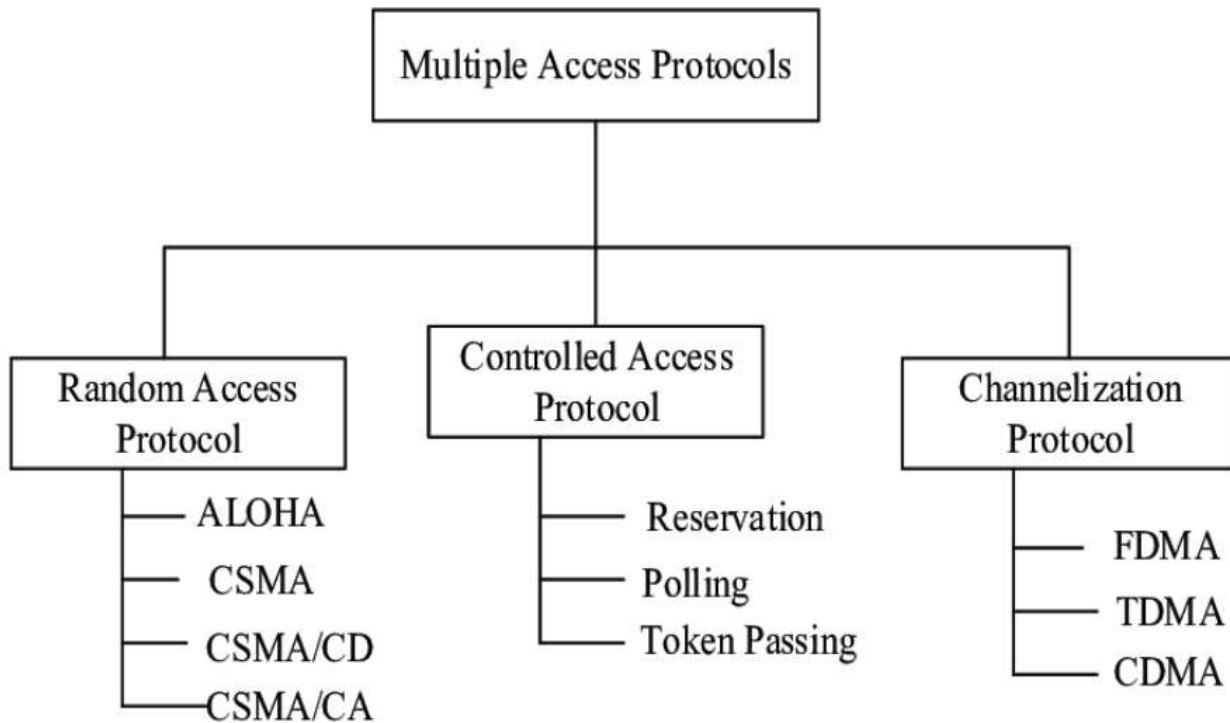
What is a collision?

A network collision occurs when two or more devices on a shared network segment attempt to transmit data frames simultaneously, causing the signals to interfere and become corrupted.



WHY MULTIPLE ACCESS PROTOCOLS?

If there is a dedicated link between the sender and the receiver then a data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously. Hence multiple access protocols are required to decrease collision and avoid Crosstalk.



1 RANDOM ACCESS PROTOCOLS

- In this, all stations have the same superiority, that is no station has more priority than another station.
- Any station can send data depending on the medium's state(idle or busy).
- In a Random access method, each station has the right to the medium without being controlled by any other station.
- If more than one station tries to send, there is an access conflict (COLLISION) and the frames will be either destroyed or modified.
- To avoid access conflict, each station follows a procedure.
 1. When can the station access the medium?
 2. What can the station do if the medium is busy?
 3. How can the station determine the success or failure of the transmission ?
 4. What can the station do if there is an access conflict ?
- **Types**
 - ALOHA
 - Slotted ALOHA
 - CSMA/CD
 - CSMA/CA

2 CONTROLLED ACCESS PROTOCOLS

- In controlled access, the stations consult one another to find which station has the right to send.
- A station cannot send unless it has been authorized by other stations.
- Types
 - Reservation
 - Polling
 - Token Passing

3 CHANNELIZATION PROTOCOLS

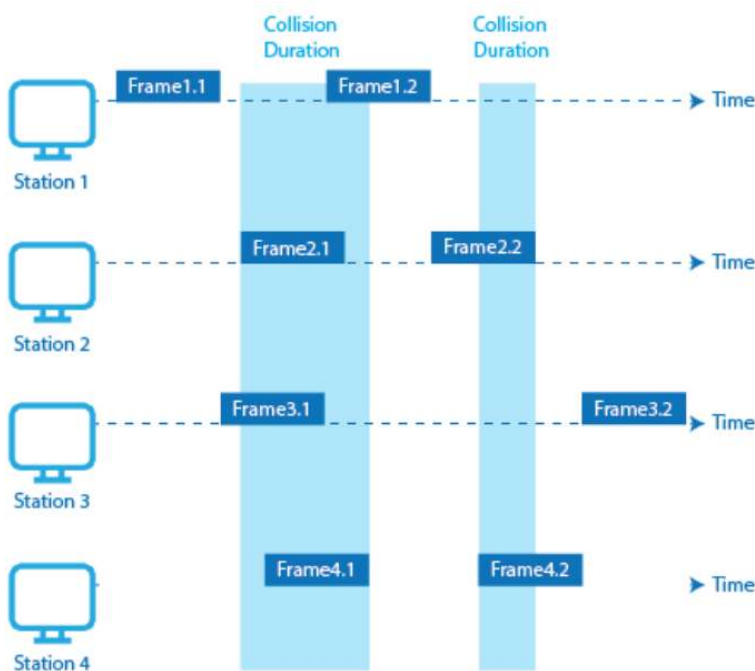
- Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations.
- Types
 - FDMA
 - TDMA
 - CDMA

ALOHA

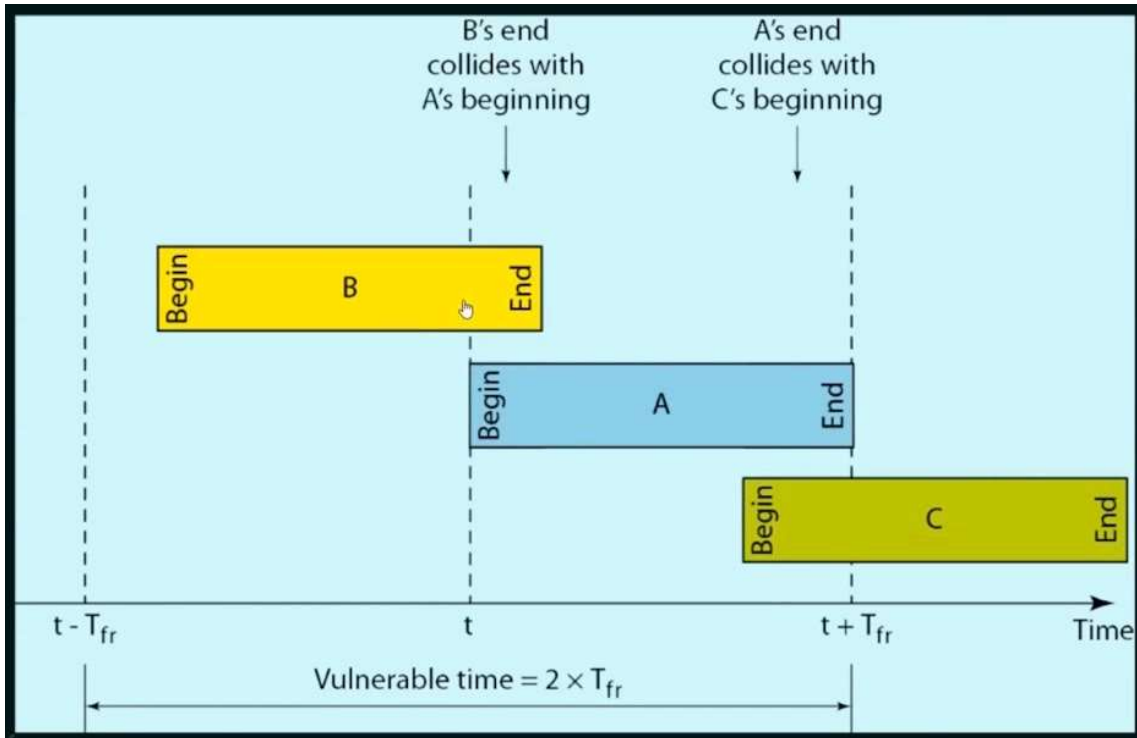
- Aloha is a random access protocol.
- It was actually designed for WLAN but it is also applicable for shared medium.
- In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.
- Types:
 - Pure Aloha
 - Slotted Aloha

PURE ALOHA

- Pure ALOHA allows stations to transmit whenever they have data to be sent.
- When a station sends data it waits for an acknowledgement.
- If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time (T_b) and re-sends the data.
- Since different stations wait for different amounts of time, the probability of further collision decreases.
- The throughput of pure aloha is maximized when frames are of uniform length.

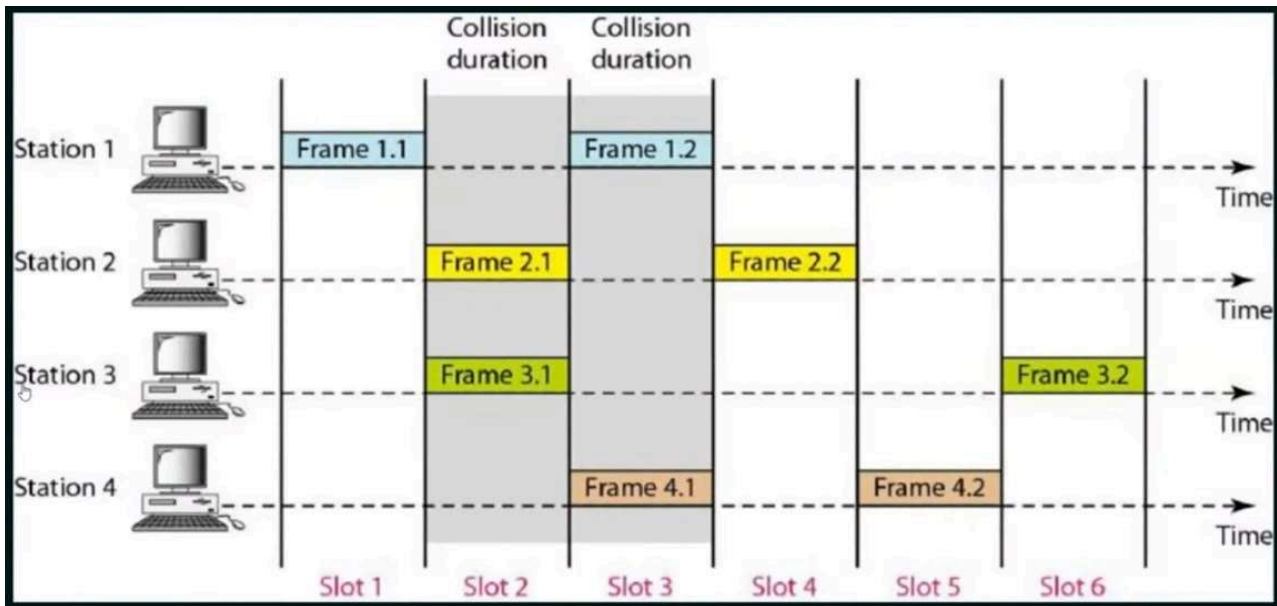


- Whenever two frames try to occupy the channel at the same time, there will be a collision and both will be garbled.
- If the first bit of a new frame overlaps with just the last bit of a frame almost finished, both frames will be totally destroyed and both will have to be retransmitted later.
 - Vulnerable Time = $2 \times T_{fr}$

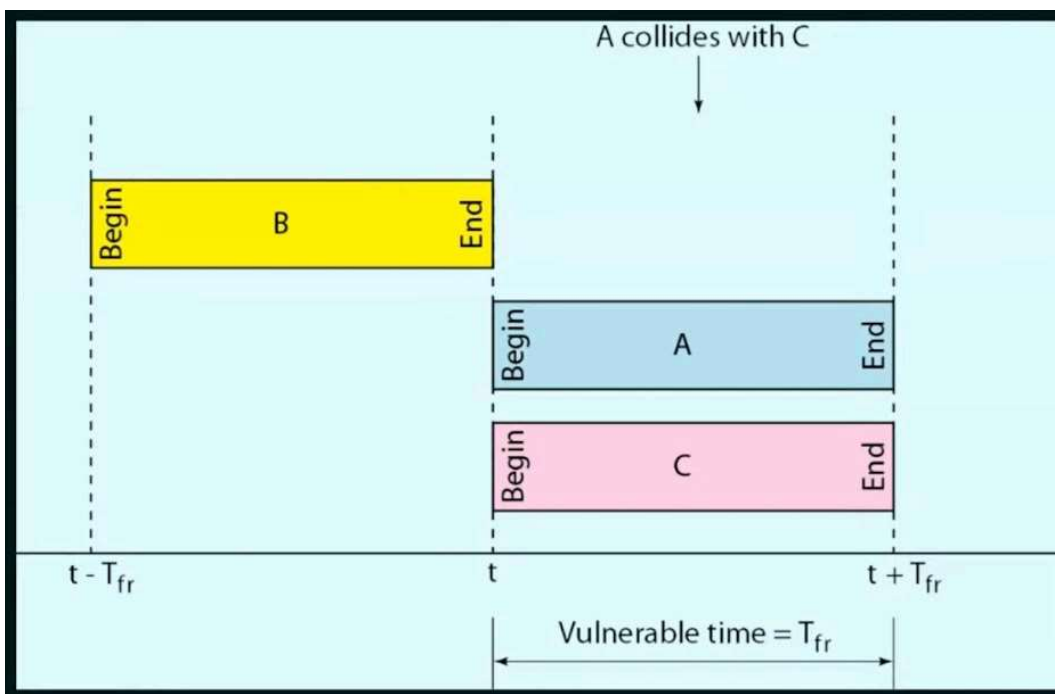


Slotted Aloha

- It was developed just to improve the efficiency of pure aloha as the chances for collision in pure aloha are high.
- The time of the shared channel is divided into discrete time intervals called slots.
- Sending of data is allowed only at the beginning of these slots.
- If a station misses out the allowed time, it must wait for the next slot. This reduces the probability of collision.

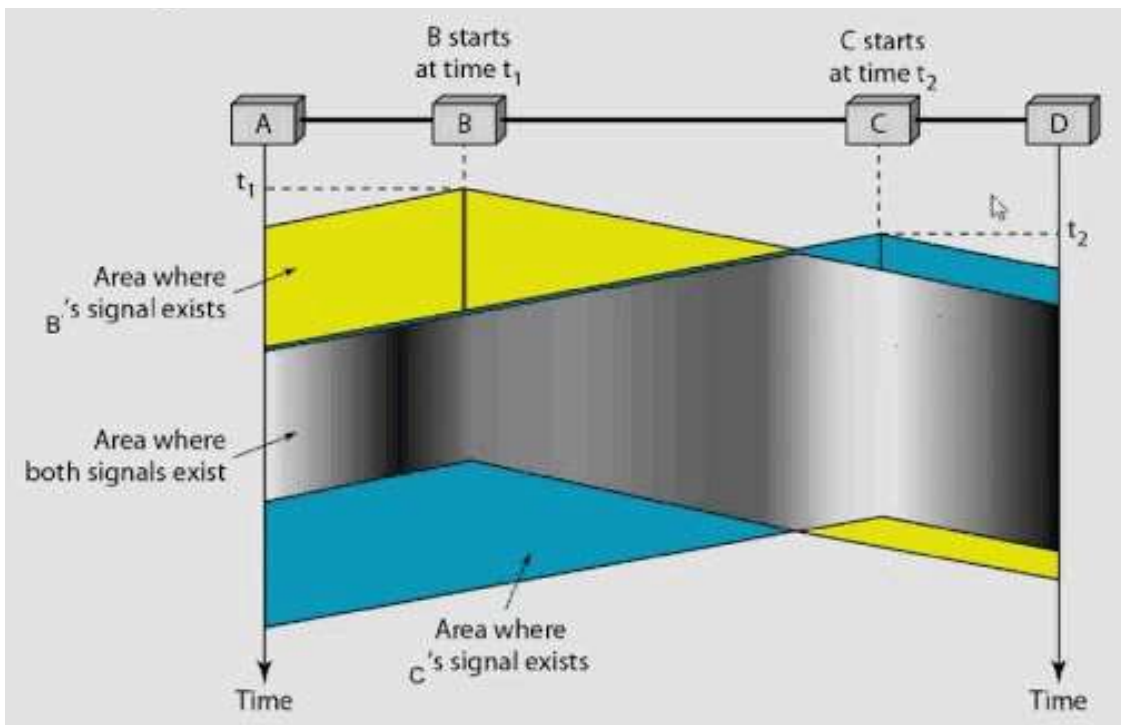


- **Vulnerable Time**= Frame Transmission Time.



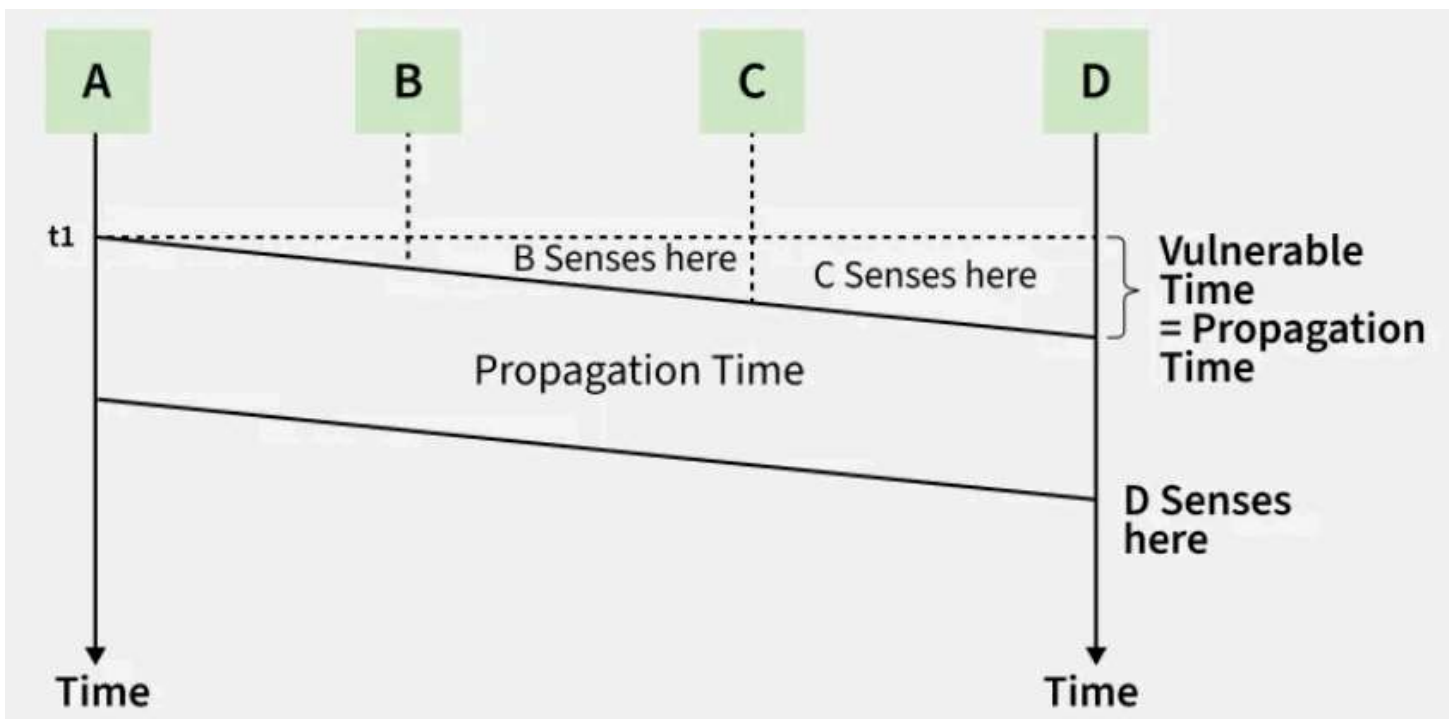
Carrier Sense Multiple Access (CSMA)

- To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed. The chance of collision can be reduced if a station senses the medium before trying to use it.
- CSMA requires that each station first listen to the medium (or check the state of the medium) before sending, so "sense before transmit". CSMA can reduce the possibility of collision, but it cannot eliminate it.
- The possibility of collision still exists because of propagation delay; when a station sends a frame, it still takes time for the first bit to reach every station and for every station to sense it.
- In other words, a station may sense the medium and find it idle, only because the first bit sent by another has not yet been received.
- At time t_1 station B senses the medium and finds it idle, so it sends a frame. At time t_2 ($t_2 > t_1$) station C senses the medium and finds it idle because, at this time, the first bits from station B have not reached station C. Station C also sends a frame. The two signals collide and both frames are destroyed.



Vulnerable Time

- The vulnerable time for CSMA is the **propagation time** T_p .
- When a station sends a frame and any other station tries to send a frame during this time, a collision will result.
- But if the first bit of the frame reaches the end of the medium, every station will already have heard the bit and will refrain from sending.
- Station A has sent a frame at time t_1 which reaches the rightmost station D, at time $t_1 + T_p$



Persistence Methods

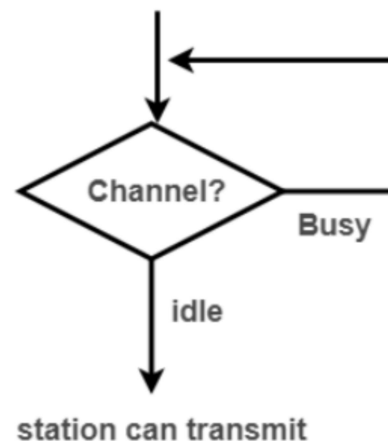
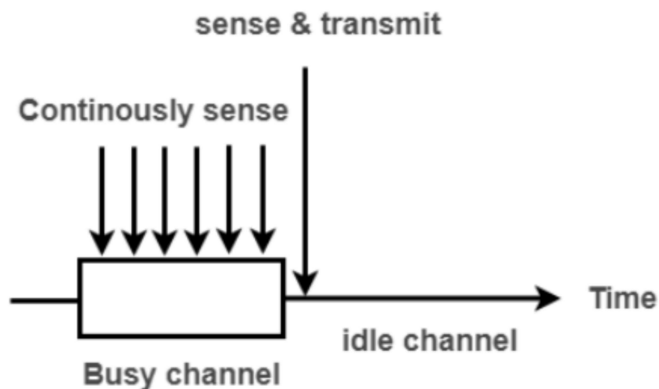
What should a station do if the channel is busy? What should a station do if the channel is idle?

Three methods have been given to answer these questions:

- 1-persistent method
- Non-persistent method
- P-persistent method.

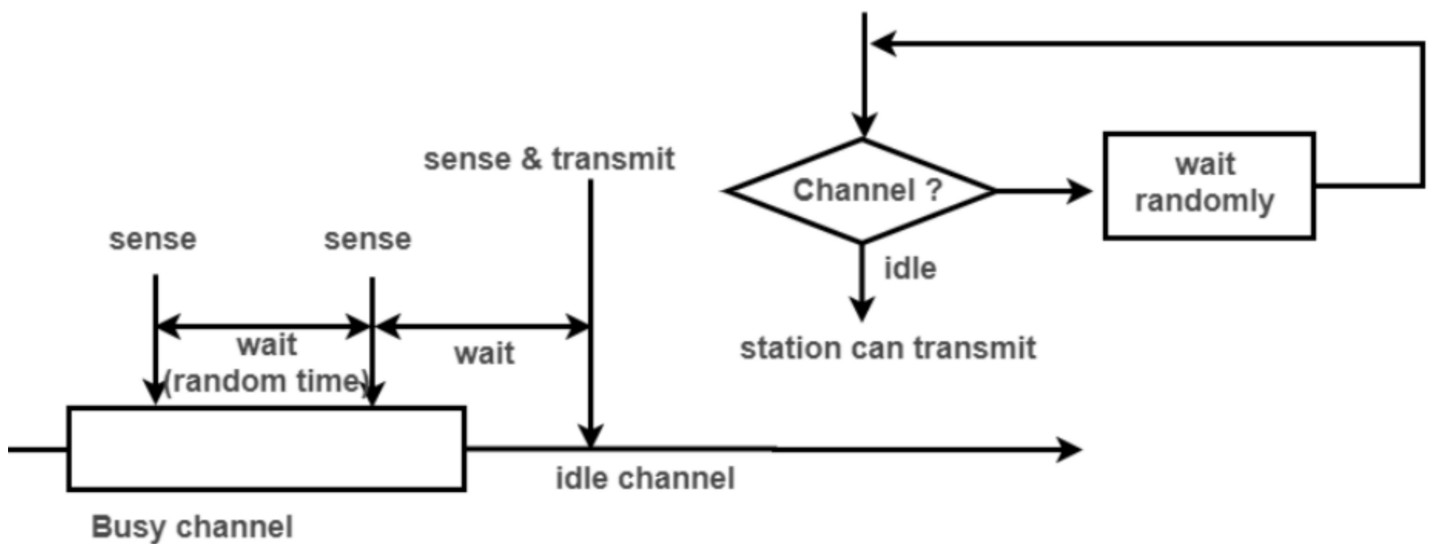
1-Persistent

The 1-persistent method is simple and straightforward. In this method, after the station finds the line idle, it sends its frame immediately (with probability 1). This method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately.



Nonpersistent

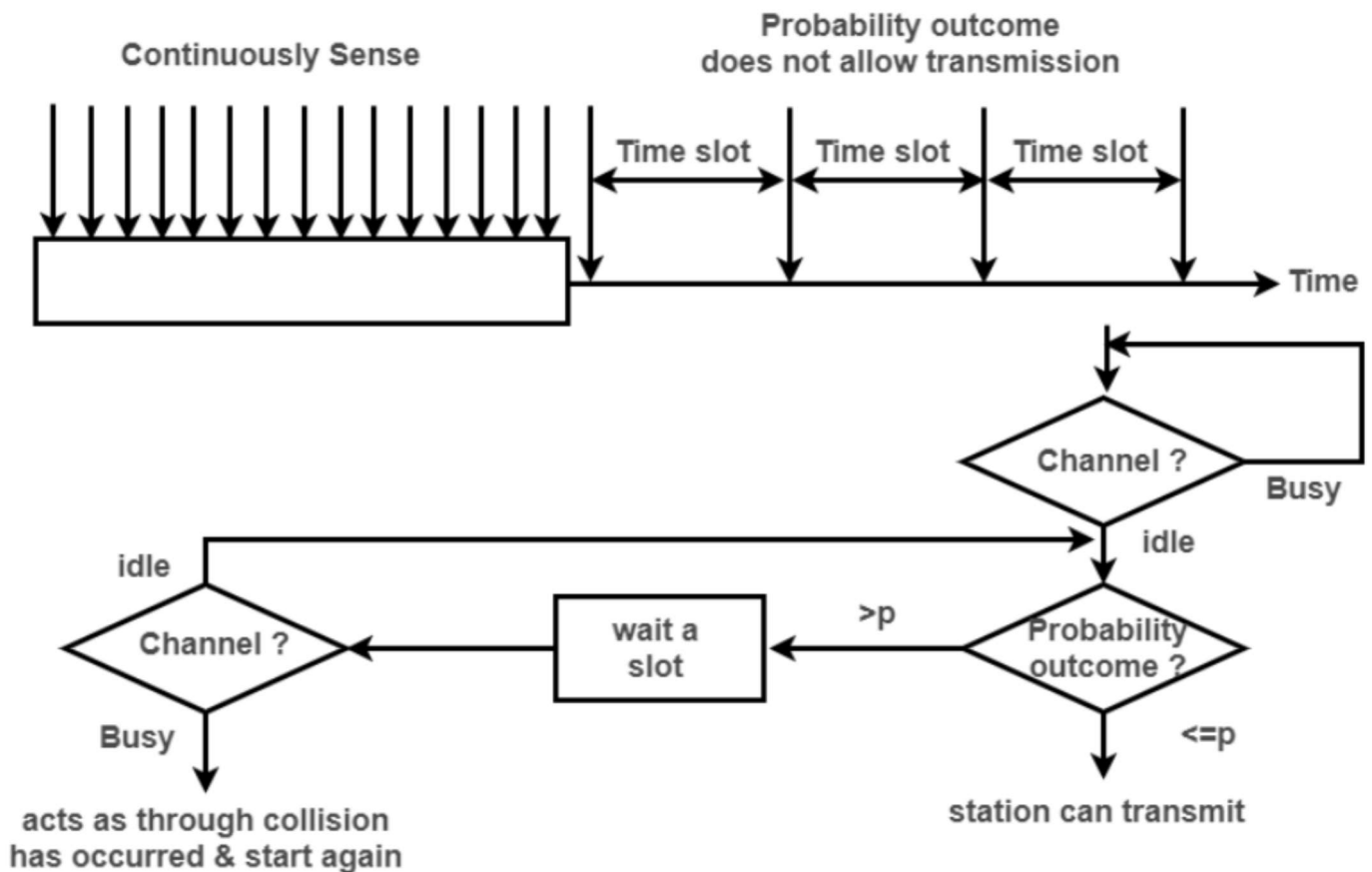
- In the nonpersistent method, a station that has a frame to send senses the line. If the line is idle, it sends immediately. If the line is not idle, it waits a random amount of time and then senses the line again.
- The nonpersistent approach reduces the chance of collision because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously. However, this method reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send.



P-Persistent

The p-persistent approach combines the advantages of the other two strategies. It reduces the chance of collision and improves efficiency. In this method, after the station finds the line idle it follows these steps:

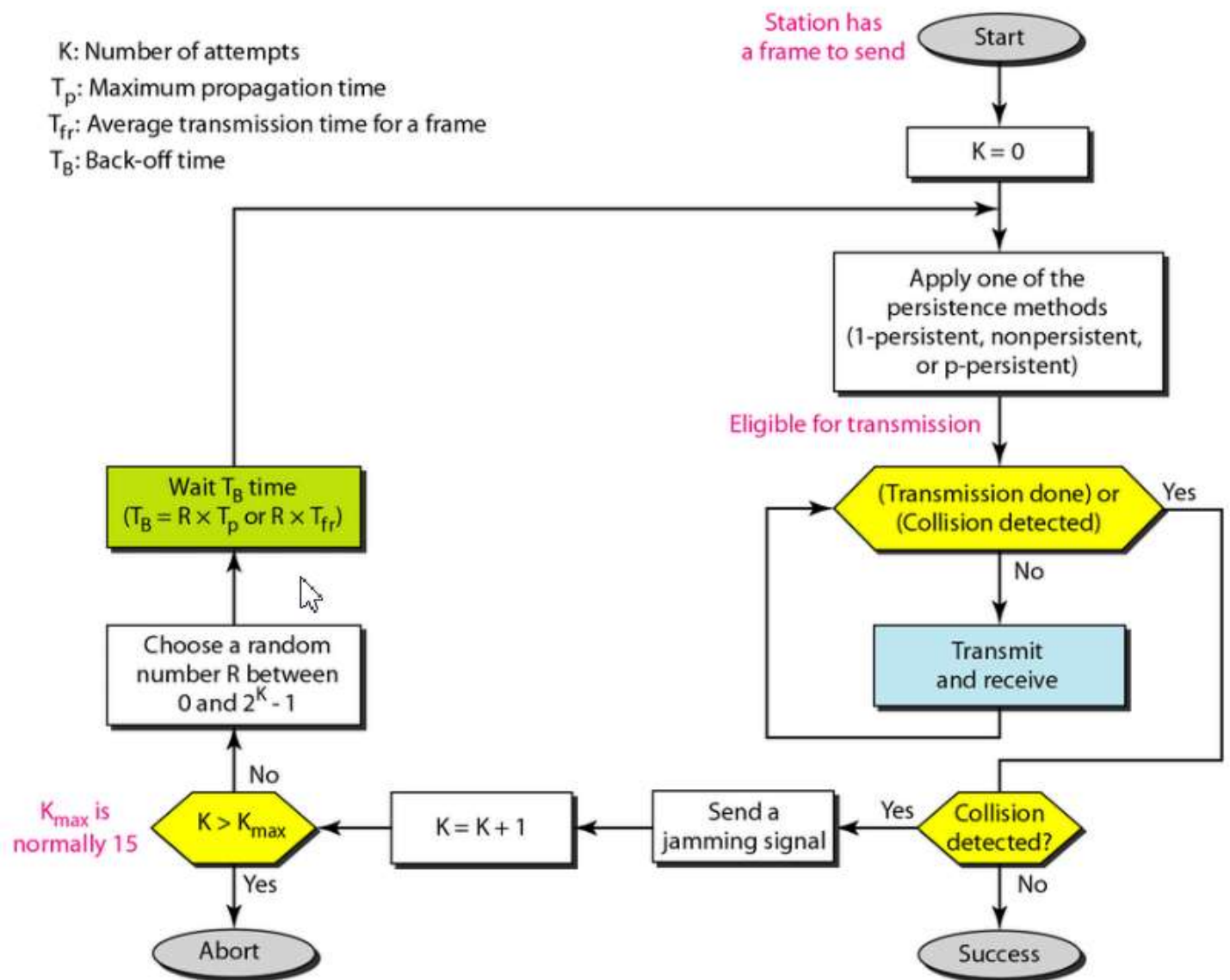
- With probability p , the station sends its frame.
- With probability $q = 1 - p$, the station waits for the beginning of the next time slot and checks the line again.
- If the line is idle, it goes to step 1.
- If the line is busy, it acts as though a collision has occurred and uses the backoff procedure.



CSMA/CD

Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however, there is a collision, the frame is sent again.
- Minimum Frame Size - For CSMA / CD to work, we need a restriction on the minimum frame size. Before sending the last bit of the frame, the sending station must detect a collision, if any, and abort the transmission.
- This is so because the station, once the entire frame is sent, does not monitor the line for collision detection. Therefore, the frame transmission time T_{fr} must be at least two times the maximum propagation time T_p
- To understand the reason, let us think about the worst-case scenario. If the two stations involved in a collision are the maximum distance apart, the signal from the first takes time T_p to reach the second, and the effect of the collision takes another time T_p to reach the first. So the requirement is that the first station must still be transmitting after $2T_p$

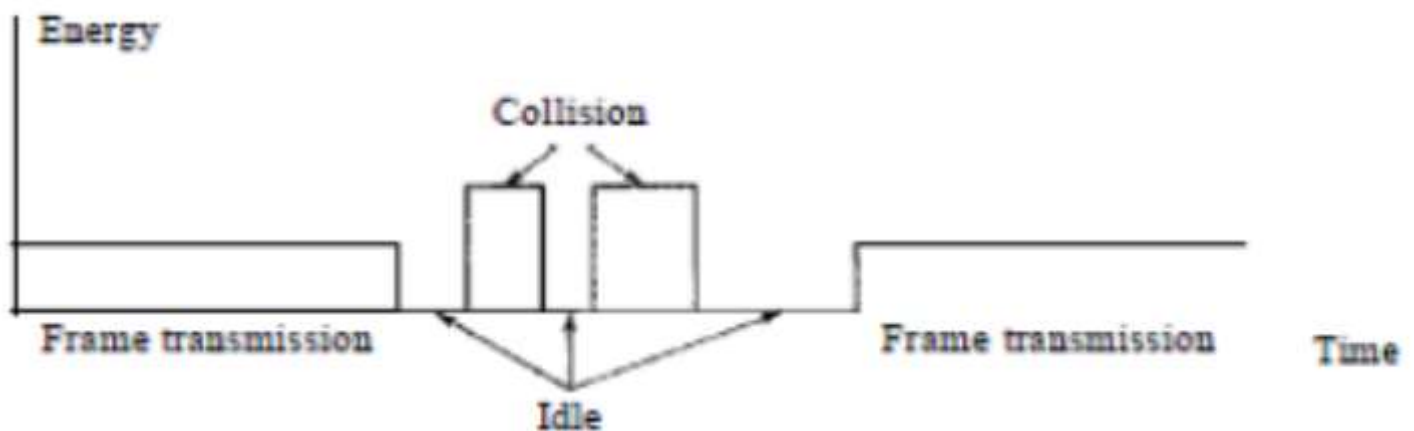


Explanation:

- The station that has a ready frame sets the back off parameter to zero.
- Then it senses the line using one of the persistent strategies.
- It then sends the frame. If there is no collision for a period corresponding to one complete frame then the transmission is successful.
- Otherwise the station sends the jam signal to inform the other stations about the collision.
- The station then increments the back off time and waits for a random back off time and sends the frame again.
- If the back off has reached its limit then the station aborts the transmission.
- CSMA/CD is used for traditional Ethernet.

Energy Level

- We can say that the level of energy in a channel can have three values: zero, normal, and abnormal. At the zero level, the channel is idle. At the normal level, a station has successfully captured the channel and is sending its frame.
- At the abnormal level, there is a collision and the level of the energy is twice the normal level. A station that has a frame to send or is sending a frame needs to monitor the energy level to determine if the channel is idle, busy, or in collision mode.



Carrier Sense Multiple Access with Collision Avoidance

CSMA/CA appropriately organizes the data transmissions by regulating the communication over a computer network with a common medium of transfer. CSMA/CA keeps checking for any transmissions from the other stations (nodes) in the channel (shared medium). It starts transmitting once the channel is free of any transmissions. **The moment a collision is detected by a station, the transmission is halted and a jam signal is sent by the station, which then waits a certain time duration before it retransmits.**

CSMA/CA avoids collisions using the following 3 core techniques:

1. Interframe Space

The station does not immediately transmit on finding the channel to be clear. Rather, it waits for a certain fixed amount of time referred to as I.F.S. (Interframe Space). The reason is that there is a possibility of transmissions having been started from the very same distant station, and the signal from that station has not yet reached the other stations.

Hence, **the aim of using IFS time is to enable this transmitted signal to reach the other stations.** After the IFS time, if the channel stays idle, the station may send. But it must still wait for a time duration equivalent to the contention time. The priority of a frame or a station can also be defined using the IFS variable.

2. Contention Window

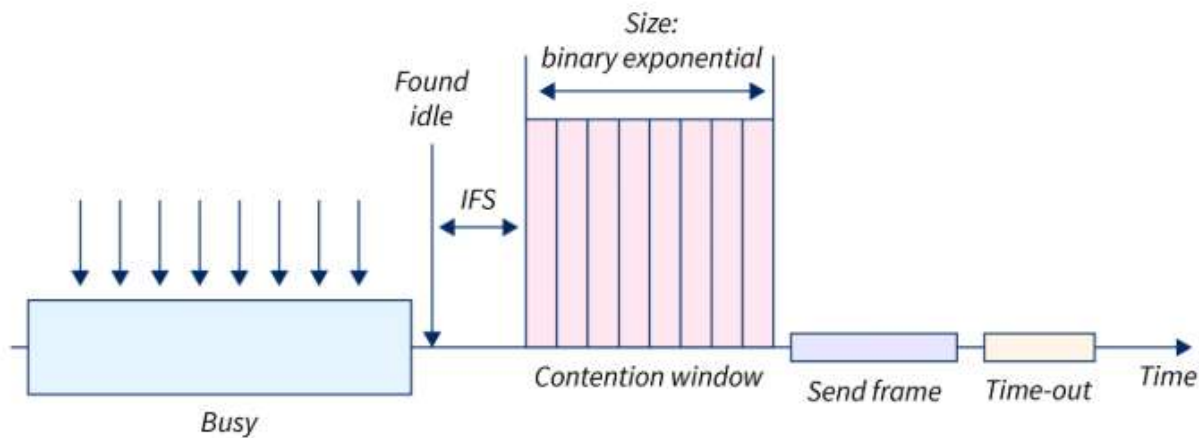
The contention window is a duration of time divided into slots/segments. Any station ready to transmit picks as its wait time, a random number of segments. This number of slots in the contention window alters as per the binary exponential back-off strategy, meaning that the initial value set is 1, doubling every time the station can't detect a free channel after the IFS time.

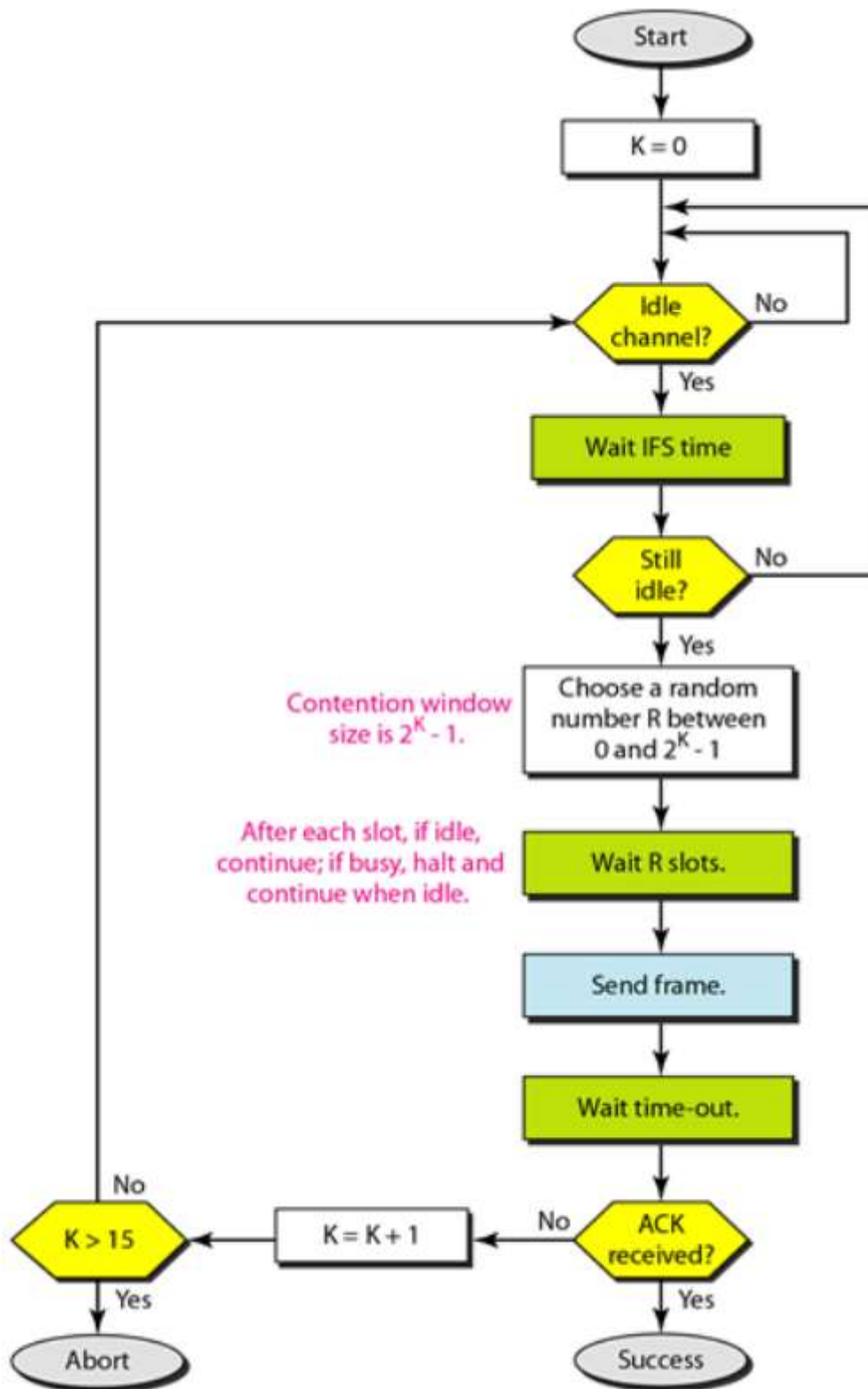
In a contention window, the station is required to check the channel at the end of every time slot. If the channel is found to be busy by the station, the process need not be

restarted. The station just halts the timer and resumes it once the channel is determined to be cleared.

3. Acknowledgments

Even after having taken so many precautions, there still exists a chance of collisions occurring and destroying the data. The time-out timer as well as the positive acknowledgment may help ensure that the receiver receives the frame.





Controlled Access Protocols in Computer Network

Controlled Access Protocols (CAPs) manage how multiple devices share a common communication medium. These protocols control the right to transmit in such a way that collisions are avoided, efficiency is maintained, and fair access is ensured.

Note: Unlike random access methods (e.g., ALOHA, CSMA), in controlled access, stations do not transmit at will. Instead, they follow a systematic method to determine which station has the right to send at a given time.

What is Controlled Access?

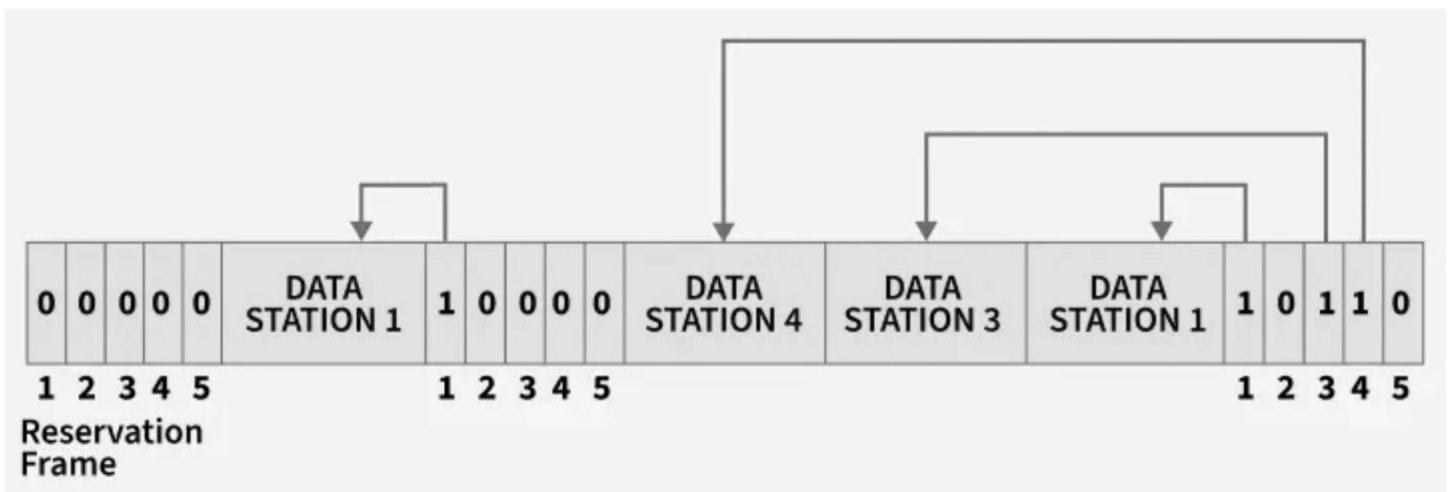
In Controlled Access, before transmitting data, a station must first obtain permission to access the channel. **At any given time, only one node can send data**, preventing collisions. This coordination is achieved using one of three methods:

- Reservation
- Polling
- Token Passing

Reservation

In this method, stations reserve slots in advance before transmission. The timeline has two parts:

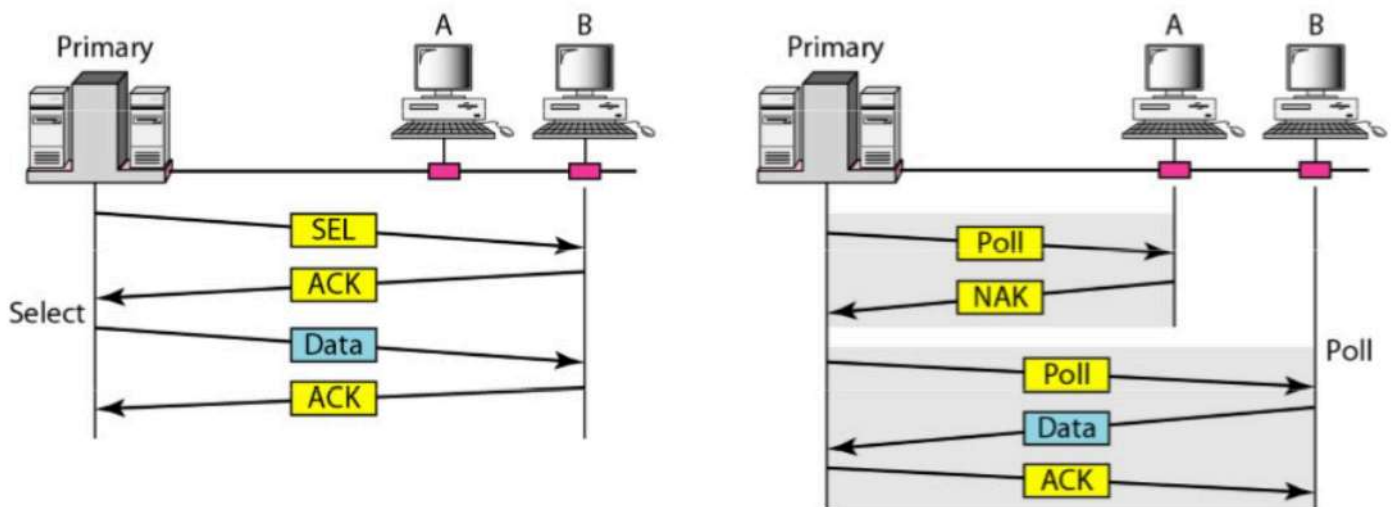
- Reservation Interval (fixed length): divided into slots, one per station.
- Data Transmission Period (variable length): reserved stations send data in order.



Example: If there are M stations, the reservation interval is divided into M slots. Each station signals intent by inserting a bit into its slot. Then, only the reserved stations transmit in the data phase.

Polling

Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations. All data exchanges must be made through the primary device even when the ultimate destination is a secondary device. The primary device controls the link; the secondary devices follow its instructions. It is up to the primary device to determine which device is allowed to use the channel at a given time. The primary device, therefore, is always the initiator of a session. If the primary wants to receive data, it asks the secondaries if they have anything to send; this is called a poll function. If the primary wants to send data, it tells the secondary to get ready to receive; this is called the select function.



Select: The select function is used whenever the primary device has something to send. Remember that the primary controls the link. If the primary is neither sending nor receiving data, it knows the link is available. If it has something to send, the primary device sends it. What it does not know, however, is whether the target device is prepared to receive. So the primary must alert the secondary to the upcoming transmission and

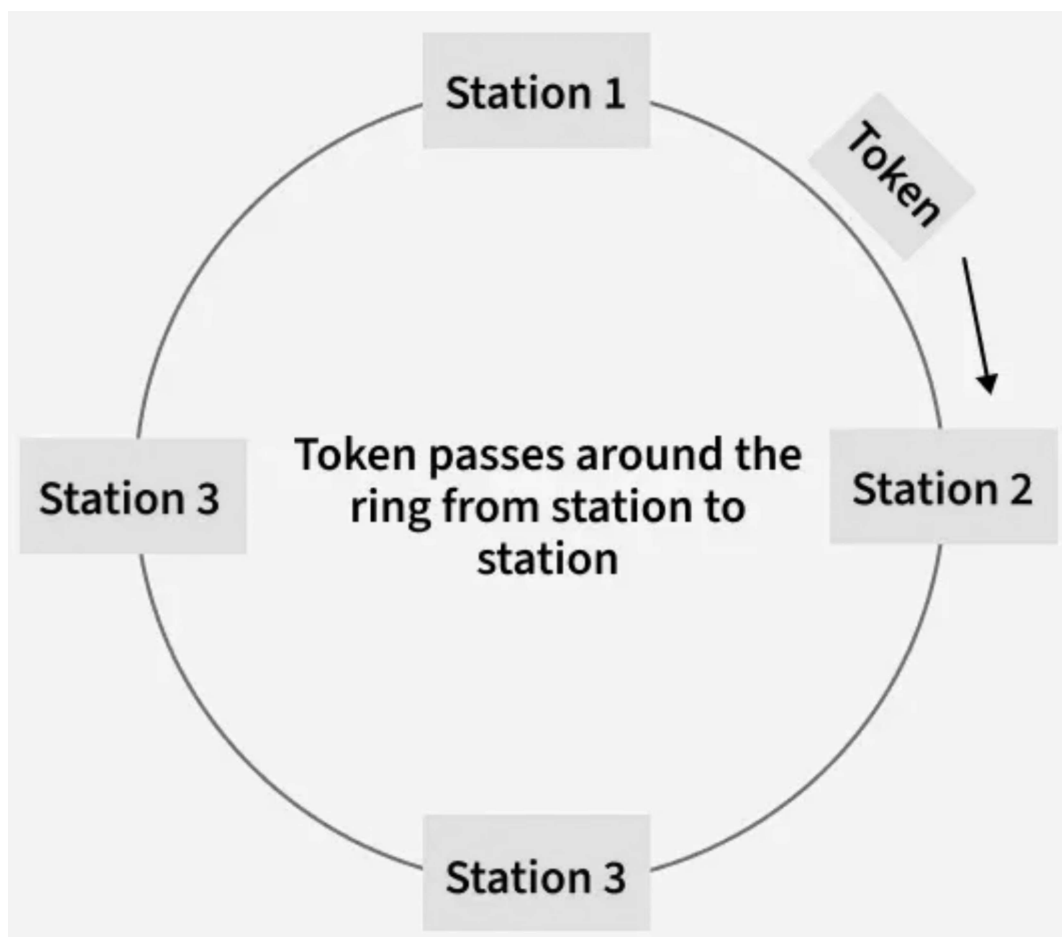
wait for an acknowledgment of the secondary's ready status. Before sending data, the primary creates and transmits a select (SEL) frame, one field of which includes the address of the intended secondary.

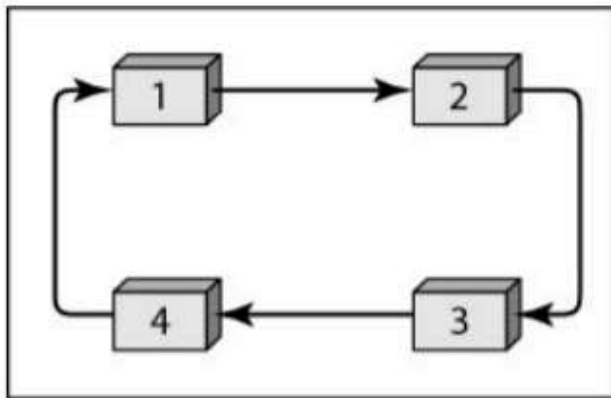
Poll: The poll function is used by the primary device to solicit transmissions from the secondary devices. When the primary is ready to receive data, it must ask (poll) each device in turn if it has anything to send. When the first secondary is approached, it responds either with a NAK frame if it has nothing to send or with data (in the form of a data frame) if it does. If the response is negative (a NAK frame), then the primary polls the next secondary in the same manner until it finds one with data to send. When the response is positive (a data frame), the primary reads the frame and returns an acknowledgment (ACK frame), verifying its receipt.

Token Passing

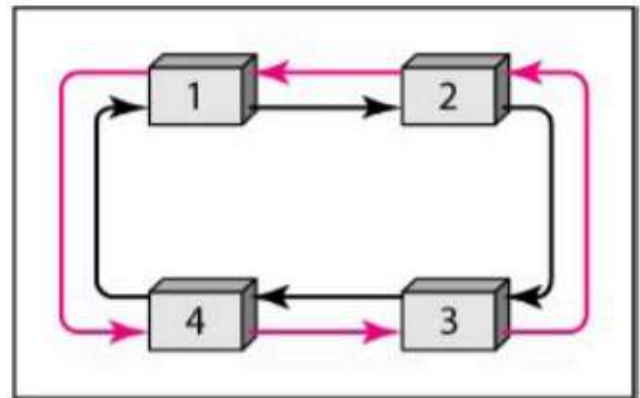
In Token Passing, stations are logically connected (ring or bus). A special frame called a token circulates among stations:

- The token represents permission to transmit.
- A station can send data only when it holds the token.
- If no data is ready, the token is simply passed on.

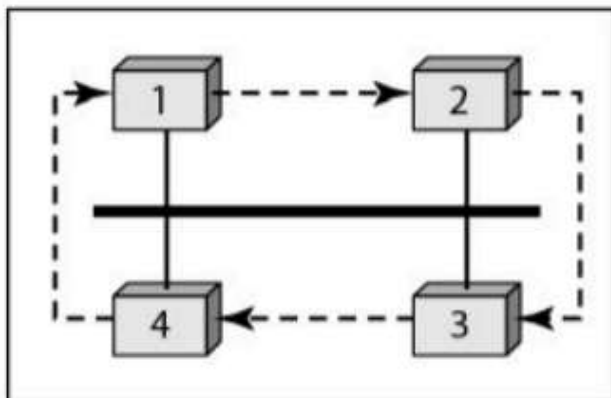




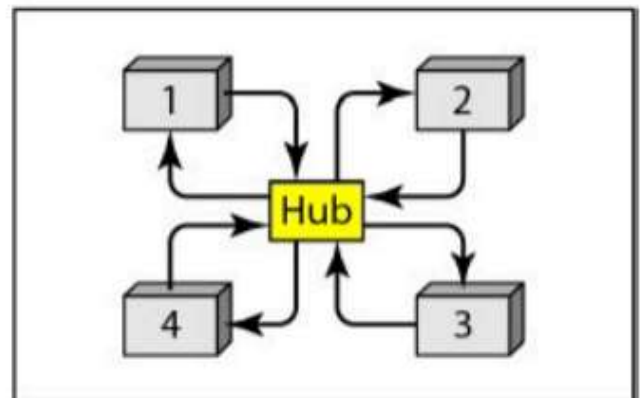
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

Common Implementations

- **Token Ring** – token passed in a ring topology.
- **Token Bus** – stations use a bus medium but pass the token in a predefined logical order.

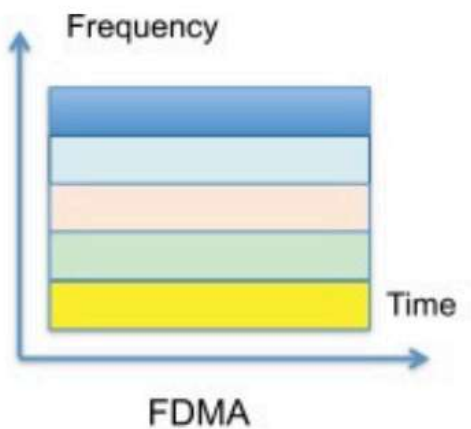
CHANNELIZATION

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols:

1. Frequency-Division Multiple Access (FDMA)
2. Time-Division Multiple Access (TDMA)
3. Code-Division Multiple Access (CDMA)

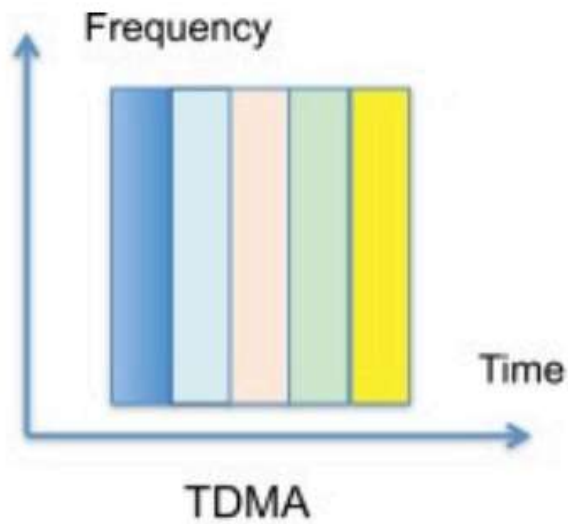
Frequency-Division Multiple Access (FDMA)

In frequency-division multiple access (FDMA), the available bandwidth is divided into frequency bands. Each station is allocated a band to send its data. In other words, each band is reserved for a specific station, and it belongs to the station all the time.



Time-Division Multiple Access (TDMA)

In time-division multiple access (TDMA), the stations share the bandwidth of the channel in time. Each station is allocated a time slot during which it can send data. Each station transmits its data in its assigned time slot.



Code-Division Multiple Access (CDMA)

Code-division multiple access (CDMA) was conceived several decades ago. Recent advances in electronic technology have finally made its implementation possible. CDMA differs from FDMA because only one channel occupies the entire bandwidth of the link. It differs from TDMA because all stations can send data simultaneously; there is no timesharing.

